Voice Receptionist - Complete Workflow Architecture

Comprehensive system design showing all components, data flows, and integrations

Input Sources

Browser Mic (Local Test)

16kHz PCM audio from /voice-test sandbox. No Twilio needed.

Twilio Voice

Production phone calls via Media Streams WebSocket. Mapped by phone number → clinic.

Audio Format

16-bit linear PCM, 1 channel, 16,000 Hz (input) / 24,000 Hz (output)

Call Initiation & Setup

Connection Routing

Twilio webhook → /twilio/voice → media_stream() WebSocket handler

Clinic Resolution

ClinicService.resolve_by_twilio_number() maps 'To' number to clinic + doctors

CallSession Creation

Per-call isolation: call_id, clinic_id, selected_doctor, STT failure counter, status

Preflight Checks

Validates Sarvam, Gemini, appointment provider before call starts

✓ Sarvam STT
API key validated

✓ Sarvam TTS
Speaker availability checked

✓ Gemini LLM
Model availability verified

✓ Appointment Provider
CruDoc connectivity confirmed

Pipecat Audio Pipeline (Per Call)

→ **MediaStreamInputTransport** — Receives binary audio frames from Twilio/browser

→ **Sarvam STT Service** — Converts audio to text (async), handles silence detection

→ **Gemini LLM Service** — Processes transcript, generates response, invokes tools

→ **Tool Executor** — Runs one of 4 tools in context of this call's clinic

→ **Sarvam/OpenAI TTS Service** — Converts LLM response to 24kHz PCM audio

→ **MediaStreamOutputTransport** — Streams TTS audio back to caller

→ **HardFallbackMonitor** — Counts STT failures; transfers if threshold hit

Gemini Tool Calls (4 Tools Available)

list_doctors

Returns clinic's doctors with specialization & external_provider_id

check_availability

Fetches real open slots from CruDoc for doctor on given date. Raises error if unreachable.

book_appointment

Books appointment with patient name, slot, reason. Returns success bool + alternatives on collision.

transfer_to_receptionist

Routes call to clinic's fallback human receptionist numbers (tried in order)

Tool Execution Safety & Isolation

- **ToolExecutor** bound to one CallSession — prevents cross-tenant access
- Each tool only sees this call's clinic_id, doctor_id, selected_doctor
- No fabricated responses — every handler uses real provider data or raises error
- Error boundaries — tool failures never crash the call, return user-friendly error

🎯 Appointment Booking - Concurrency & Consistency

Layer 1: Redis Lock — AppointmentService acquires lock on (clinic_id, doctor_id, slot). Narrows race window for concurrent calls.

Layer 2: Real Availability — Provider call (CruDoc) is the source of truth. Never cached. CruDoc responds 409 if slot occupied.

Layer 3: DB Constraint — UNIQUE(doctor_id, slot_start) on appointments table. Final guarantee: collision → IntegrityError → unwind booking + fetch alternatives

Layer 4: Fresh Alternatives — On collision, call _alternatives() for fresh availability. Failure there never masks booking error.

🏥 CruDoc Integration (Appointment Provider)

Provider Adapter Pattern

- get_provider() caches adapters by config signature
- Each call creates new httpx.AsyncClient connection
- Credential rotation → different cache key → fresh adapter
- Prevents connection pool leaks

CruDoc Endpoints Called

- GET /api/appointments/availability (check_availability)
- POST /api/appointments (book_appointment)
- Timezone passed for slot calculations
- All calls include clinic API key from .env

☎️ Twilio Integration (Production Calls)

Media Streams WebSocket

Long-lived connection per call. Binary PCM + JSON events (SID, etc.)

Signature Validation

Webhook authenticity verified using TWILIO_AUTH_TOKEN

Phone Number Mapping

Twilio 'To' number (E.164) → phone_numbers table → clinic + line config

Call Hangup & Transfer

TwiML.Dial() to transfer; Twilio API to hangup after session

Database Schema & State Management

- **Clinics**

`clinic_id (UUID) | name | timezone | ai_receptionist_enabled | appointment_provider | config (JSON)`

- **Doctors**

`doctor_id | clinic_id | name | specialization | external_provider_id (for CruDoc bookings)`

- **Phone Numbers (Lines)**

`number (E.164) | clinic_id | label | doctor_ids (JSON) | greeting | voice_id | ai_instructions | fallback_receptionists`

- **Appointments**

`appointment_id | clinic_id | doctor_id | caller_name | caller_phone | slot_start | slot_end | call_id (FK) | reason | created_at`

- **Calls**

`call_id (UUID) | clinic_id | twilio_call_sid | caller_phone | status (booked/transferred/no_action) | duration | appointments_booked (count) | started_at | ended_at`

- **Receptionists**

`phone_number | human_receptionist_numbers (for transfer fallback)`

Storage Options:

Default: SQLite

Local dev: voice_receptionist.db. Auto-created on startup. Simple, no deps.

PostgreSQL

Production: DATABASE_URL env var. AsyncPG driver. Async support.

Redis (Optional)

REDIS_URL env var. Used for appointment booking locks in multi-process deployments.

Core Services

ClinicService

resolve_by_twilio_number() → ResolvedClinic (immutable per call). Prevents tenant drift.

AppointmentService

check_availability(), book_appointment(), _alternatives(). Handles all concurrency layers.

TransferService

transfer_call() → tries fallback receptionist numbers in order. Updates call status to TRANSFERRED.

CallService

Logs call start/end. Tracks booking count, duration, outcome for analytics.

Voice & LLM Configuration

TTS Voices (Sarvam)

- Meera (default)
- Other Sarvam speakers per config
- Clinic or line-specific voice_id override

LLM Model & Prompts

- Gemini 3.5 Flash (current, not retired 2.0)
- System prompt in prompts.py
- Tool declarations (ToolsSchema)

- Preflight validates speaker availability
- Context: clinic name, doctors, timezone

👑 Admin & Super Admin Integration

Super Admin Integration Flow

- **Super Admin App** (ReceptionistLineService) → calls Cloud Function
- **Cloud Function** (manageReceptionistLines in functions-admin) → validates role, audit-logs
- **Voice Service Admin API** (PUT /api/admin/receptionist-lines/{+E164}) → updates clinic/line/doctors
- **Authentication:** x-api-key header (VOICE_BOT_API_KEY)

Admin Endpoints

- GET /api/admin/status → Health + provider checks (optional deep=true for real calls)
- GET /api/admin/calls → Recent calls by clinic/date/limit (call outcome breakdown)
- GET /api/admin/appointments → Appointments booked by AI by clinic/date
- PUT /api/admin/clinics/{external_id}/entitlement → Toggle ai_receptionist_enabled

🧪 Browser Test Sandbox (/voice-test)

Audio Bidirectional

Browser mic (getUserMedia) → WS → Sarvam STT → Gemini → Sarvam TTS → browser speaker

WebSocket Connection

ws://localhost:8000/voice-test/ws. Runs full pipeline in-process, no Twilio.

Live Transcript

User + AI turns rendered in real time. Waveform visualization. Session state badge.

Diagnostic Logs

Preflight, WebSocket, STT, Gemini, tool execution, appointment booking traced

Local Test Limitations (Can Enable)

- ❌ LOCAL_VOICE_TEST_ALLOW_BOOKING=false → book_appointment returns error
- ❌ LOCAL_VOICE_TEST_CLINIC_ID env var → fixed test clinic (no Twilio mapping)
- ❌ Transfers disabled → transfer_to_receptionist returns error
- ✅ All STT/LLM/TTS live if API keys set in .env

Logging & Monitoring

Structured Logging

JSON logs via structlog. Components: startup, voice_tools, appointment_service, pipeline

Call Audit Trail

calls table tracks: call_id, clinic_id, caller_phone, status, appointments_booked, duration